A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Importance & features of energy resources - Set 1

Mark scheme

AQA spec 3.3.1-3.3.2; Energy resources; textbook pp. 170-182

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) **1 But not proportional:** B has a high GDP but much lower energy use than A or E – C uses relatively high energy for its GDP **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes **1** Manufacture and transport of the extra products bought uses energy (indirect use) – larger houses use more energy to build **1** Affluent countries can afford energy for non-essential uses – cheap local energy (eg petrol in the USA) encourages wasteful use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Higher energy prices / taxes encouraging conservation **1** Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness **1** Milder climate needing less heating or air conditioning **1 Service-based economy:** heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted **1** Smaller / more efficient vehicles, better public transport, better insulated buildings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Decline of heavy industries such as steel and aluminium smelting in the UK **1** The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country **OR** efficiency improvements **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.5	1 mark for the point below. **1 Water:** pumping, treatment / sterilisation, sewage treatment – Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** The amount of energy contained in a given mass of the resource (or harnessed per kg of equipment for renewables) **1** High energy density means smaller quantities are needed so storage and transport are easier and cheaper (eg 75 t of aviation fuel carries a 400 t aircraft 5600 km) **1** High energy density fuels reach high temperatures needed for smelting / steam turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 500 000 / 28 = 17 857 kg **1** = 17.9 tonnes (accept 18; book quotes 13 t for 0.7% enriched fuel) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The electricity they generate can power high-energy-requirement technologies **1** Electricity can be converted to high energy density secondary fuels, eg hydrogen by electrolysis, or stored (batteries, pumped storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1 Intermittent:** not available all the time, eg tidal (but the times are predictable so alternatives can be planned) **1 Unpredictable:** intermittent and the timing / amount cannot be forecast accurately, eg wind, solar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • Any appropriate named example, including ones that are not listed here. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous	2
02.5	1 mark for the point below. **1** Gaseous natural gas has a very low energy density by volume – LNG at -160 °C has 160 times the energy density so a ship can carry a worthwhile amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-178; AQA spec 3.3.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1 Renewable:** re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) **1 Depletable:** use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably **1** Wood / biofuels are renewable but depletable if forests are felled faster than they regrow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Other factors (technology, location, cost) restrict how much can be harnessed **1 Examples:** coal too deep to mine – winds high above the ground / far offshore where turbines cannot be placed – waves in deep water where equipment cannot be anchored – low-intensity sunlight that cannot produce high temperatures **1** New technologies or higher prices may make these usable in future **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1 Wind:** strong reliable winds - shallow seas, open plains, uplands – low land-use conflict (not near urban areas, ecologically sensitive or scenic areas) – near the grid **1 HEP:** high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock **1 Solar:** high light intensity, low cloud cover **1 Wave:** coasts with strong reliable prevailing winds and a long fetch **1 Tidal:** large tidal range, coastal features (estuaries, channels) that focus the flow **1 Geothermal:** hot rocks near the surface / recent volcanic activity **1 Biofuels:** nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **HEP** → electricity generated from falling water • **impermeable** → does not let water through **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Solar PV cost fell dramatically from about 480 to 50 $/MWh (about 90%), the steepest fall **1** Wind fell steadily from 200 to 45 **1** Gas stayed roughly constant at 60-70 – solar and wind became cheaper than gas around 2015-2020 and are projected to fall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Development / research costs are still being paid for and there are no economies of scale in manufacture **1** They compete with established technologies whose development costs were paid long ago and which benefit from mass production and existing infrastructure **1** As output grows, mass production, improved designs / efficiency and experience reduce costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

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Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most of the renewable cost is the initial equipment, paid for with loans on which interest must be paid – running costs are very low **1** Most of the gas project's cost is running costs (fuel) paid from income without borrowing, so it has lower interest payments **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Environmental damage (acid rain damage to buildings, crops, forests; health effects of air pollution; climate change) is paid for by others (health service, agriculture) not the energy user **1** Some impacts (climate change) cannot yet be costed – decommissioning and waste costs are also uncertain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 179-181; AQA spec 3.3.2-3.3.3.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Most energy comes from non-renewable fossil fuels whose reserves are depleting – the rate of new discoveries has fallen sharply since the 1980s **1 Use causes unacceptable environmental damage:** atmospheric pollution, climate change, oil pollution, habitat loss **1 Economic / social:** depleting resources become more expensive, reducing future affluence – risk of an energy gap if new technologies are not developed in time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

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Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coal mining:** habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise **1 Oil and gas extraction:** spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO2 / CO2 / smoke **1** Pipelines causing habitat damage during construction – tanker transport risks **1** Fuel processing (refineries, coal washing) pollution – embodied energy of equipment and site development for all resources **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	4
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) **1** Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise **1** By then depletion would reduce supply faster than replacements could be built, causing shortages **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	

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Question	Marking guidance	Marks
06.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Features:** renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • **Fossil fuels:** high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO2), politically concentrated (Middle East oil) • **Nuclear:** very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • **Renewables:** low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • **Sustainability concepts:** impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • **Economic and political factors:** subsidies, guaranteed prices, planning, external costs not included in price, energy security • **Judgement:** a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable – transition must start before depletion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **base load** → the steady minimum demand that must always be met **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and CO_2, and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition.	
	Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen.	
	No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it.	
	Weaker answer - Level 1 (2 marks)	
	Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy.	
	What makes the difference	
	• Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry).	
	• Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely.	
	• The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion.	

Do not write outside the box

Question	Marking guidance	Marks
06.2	1 mark for the point below. **1** Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Do not write outside the box